

Clinical Threshold Justification

Vital Sign	Threshold	Clinical Reason
SBP < 70 mmHg	Severe hypotension	Indicates shock and poor organ perfusion
SBP > 200 mmHg	Hypertensive crisis	Increased risk of stroke or organ damage
SpO ₂ < 90%	Hypoxia	Suggests inadequate oxygenation
Pulse > 130 bpm / < 40 bpm	Tachycardia / Bradycardia	Indicates cardiac stress or reduced output
RR > 30 / < 8 breaths per min	Respiratory distress / depression	Reflects respiratory compromise
Temp > 38.5°C / < 35°C	Fever / Hypothermia	Suggests infection or severe illness
GCS < 8	Severe neurological impairment	Risk of airway compromise

Pseudocode

Adult patient triage algorithm
Designed to identify patients at risk of clinical deterioration

risk_score = 0

alerts = []

Circulation assessment

if SBP < 70:

 risk_score += 3

 alerts.append("Shock risk")

if SBP > 200:

 risk_score += 3

 alerts.append("Hypertensive crisis")

Oxygenation assessment

if SpO₂ < 90:

 risk_score += 3

 alerts.append("Hypoxia")

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# Heart rate assessment
if Pulse > 130 or Pulse < 40:
    risk_score += 2
    alerts.append("Critical heart rate")

# Respiratory assessment
if RR > 30 or RR < 8:
    risk_score += 2
    alerts.append("Respiratory distress")

# Temperature assessment
if Temp > 38.5 or Temp < 35:
    risk_score += 1
    alerts.append("Temperature abnormality")

# Neurological assessment
if GCS < 8:
    risk_score += 3
    alerts.append("Severe neurological impairment")

# Final risk classification
if risk_score >= 6:
    risk_status = "Critical"
elif risk_score >= 3:
    risk_status = "High"
elif risk_score >= 1:
    risk_status = "Moderate"
else:
    risk_status = "Normal"

print(risk_status)
print(alerts)

```

The algorithm uses a cumulative scoring system to reflect real clinical triage practice, where multiple simultaneous abnormalities indicate greater patient deterioration than a single abnormal vital sign. Each parameter was selected based on established emergency medicine principles, with higher weights assigned to life-threatening indicators such as hypoxia, severe hypotension, and reduced consciousness, and thresholds aligned with standard adult physiological ranges used in clinical settings. EDA findings supported this design, particularly the strong association between respiratory rate, oxygen saturation, and patient deterioration, which informed their prioritisation in the scoring system. This approach improves logical flow, clinical relevance, and practical feasibility by ensuring the system integrates multiple vital signs into a single structured risk decision rather than relying on isolated measurements.